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Kacharam, Shamshabad, Hyderabad – 501218, Telangana, India

## DEPARTMENT OF COMPUTER SCIENCE AND ENGINEERING (AI&ML)

# Major Project Submission (A. Y. 2026-2027)

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### 1. Project Title:

A Multimodal Generative AI System for Adaptive Crop Planning and Proactive Farm Intelligence

### 2. Team Details:

Team Number – 7

Team Members:

Edara Bharath Kalyan Reddy – 23881A6618

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S. S. Abhiram – 23881A6652

**3. Guide name and designation:** Mr. Manzoor Mohammad

### 4. Problem Statement:

Agricultural decision-making remains highly dependent on fragmented information sources, generalized farming recommendations, and reactive expert consultation, making it difficult for farmers to obtain timely and personalized guidance based on their specific soil conditions, crop type, geographical location, weather variations, and current crop growth stage. Existing agricultural advisory systems often address isolated tasks such as crop recommendation, soil analysis, disease detection, or question answering, but lack an integrated mechanism that continuously supports farmers throughout the complete cultivation lifecycle. Therefore, there is a need for a multimodal and context-aware agricultural knowledge assistant capable of analyzing visually inferable soil characteristics, combining them with environmental and farm-specific parameters to recommend suitable crops, generating personalized cultivation timelines, maintaining crop lifecycle context, and proactively delivering timely guidance on irrigation, fertilization, pest and disease risks, and other critical farming activities. Integrating Generative AI with Retrieval-Augmented Generation can further enable grounded, contextually relevant, and accessible agricultural assistance, thereby supporting informed, timely, and sustainable farming decisions from crop selection to harvesting.

### 5. Project Objectives:

**Objective 1:** To develop a multimodal soil analysis and crop recommendation module that analyzes visually inferable soil characteristics from uploaded images and combines them with contextual parameters such as location, season, weather conditions, and optional soil attributes to suggest potentially suitable crops.

**Objective 2:** To design a personalized crop lifecycle planning and notification system that generates crop-specific cultivation timelines and provides timely, context-aware reminders for key farming activities such as irrigation, fertilization, pest and disease monitoring, and harvesting based on crop stage and available environmental information.

**Objective 3:** To build a Generative AI-powered agricultural knowledge assistant using Retrieval-Augmented Generation (RAG) that retrieves information from curated agricultural knowledge sources and generates contextually relevant responses to farmer queries while considering available farm, crop, and lifecycle information.

## 6. Abstract:

Agriculture faces challenges such as climate variability, declining soil quality, inefficient resource utilization, and limited access to timely and personalized expert guidance. Existing agricultural advisory systems often provide generalized recommendations or address isolated tasks, lacking continuous support throughout the crop cultivation lifecycle. This project proposes a Generative AI Powered Agricultural Knowledge Assistant, a multimodal and context-aware decision-support system designed to assist farmers from crop selection to harvesting. The proposed system enables farmers to upload soil images, which are analyzed using computer vision and deep learning techniques to identify visually inferable soil characteristics. These insights are combined with contextual parameters such as location, season, weather conditions, and optional soil attributes to suggest potentially suitable crops. Once a crop is selected, the system generates a personalized cultivation timeline covering activities such as sowing, irrigation, fertilization, pest and disease monitoring, and harvesting. A crop lifecycle mechanism tracks crop age and expected growth stages to support timely, context-aware notifications. The system further integrates Generative AI with Retrieval-Augmented Generation (RAG) to retrieve information from curated agricultural knowledge sources and generate contextually relevant responses to farmer queries. By combining multimodal analysis, crop recommendation, lifecycle planning, proactive notifications, and grounded knowledge assistance, the proposed system aims to support more informed and accessible agricultural decision-making.

**Keywords:** Generative Artificial Intelligence, Multimodal AI, Precision Agriculture, Soil Image Analysis, Crop Recommendation, Crop Life Cycle Management, Retrieval Augmented Generation, Agricultural decision support.

## 7. Tools and Technologies Used:

Programming Language: Python, React.js

Libraries: Tensorflow, OpenCV, scikit-learn, LangChain.

APIs: OpenWeather API, APScheduler, GeminiAPI.

Platforms: VS code, Github.

Database: SQLite.

## 8. Expected Deliverables:

1. A functional multimodal agricultural decision-support system that integrates soil-image analysis, contextual crop recommendation, personalized crop lifecycle planning, and stage-aware notifications for key farming activities.

## 9. SDG Mapping:

| SDG No | Goal Title                              | Justification                                                                                                                                                                                                                                                                                                                                                                                                                                                                       |
|--------|-----------------------------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| 4      | Quality Education                       | The Generative AI-powered knowledge assistant acts as an accessible agricultural learning and knowledge-support platform for farmers. Through natural-language interaction and context-relevant responses, it can help farmers understand cultivation practices, crop stages, soil-related considerations, pest and disease monitoring.                                                                                                                                             |
| 9      | Industry, Innovation and infrastructure | The project applies emerging digital technologies—including Generative AI, Retrieval-Augmented Generation (RAG), computer vision, machine learning, and weather-data integration—to agricultural decision support. By integrating these technologies into a unified farmer-centric platform, the project demonstrates digital innovation aimed at improving access to intelligent agricultural services.                                                                            |
| 12     | Responsible Consumption and Production  | The system provides context-aware guidance and timely reminders for activities such as irrigation, fertilizer application, and pest monitoring. Such decision support may help farmers make more informed use of agricultural inputs and reduce unnecessary or poorly timed resource application, supporting more responsible and sustainable production practices.                                                                                                                 |
| 17     | Partnerships for the goals              | The project can serve as a technology platform that integrates agricultural knowledge from curated sources with external services such as weather data and, where applicable, information from agricultural institutions or domain experts. Its architecture can support future collaboration among farmers, researchers, educational institutions, agricultural organizations, government bodies, and technology providers for knowledge sharing and digital agricultural support. |

## 10.OBE Mapping: Program Outcomes (POs)

| PO Number | Name of the PO Targeted                    | Justification                                                                                                                                                                                    |
|-----------|--------------------------------------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| PO1       | Engineering Knowledge                      | Applies knowledge of AI, machine learning, computer vision, Generative AI, RAG, web development, databases, and APIs to develop the agricultural knowledge assistant.                            |
| PO2       | Problem Analysis                           | Analyzes the problem of fragmented agricultural information and the lack of personalized guidance based on soil characteristics, crop type, location, weather conditions, and crop growth stage. |
| PO3       | Design/Development of Solutions            | Designs an integrated system for soil-image analysis, crop recommendation, personalized crop timelines, timely farming notifications, and GenAI-based agricultural query assistance.             |
| PO4       | Conduct investigations of complex problems | Involves experimenting with soil-image classification, crop recommendation models, agricultural knowledge retrieval, and evaluation of model predictions and                                     |

|      |                                        |                                                                                                                                                                                                                      |
|------|----------------------------------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
|      |                                        | generated responses.                                                                                                                                                                                                 |
| PO5  | Engineering Tool Usage                 | Uses modern tools and technologies such as Python, React.js, FastAPI, TensorFlow/Keras, scikit-learn, Gemini API, LangChain, FAISS, weather APIs, Git, and GitHub.                                                   |
| PO7  | Ethics                                 | Considers responsible AI usage by grounding agricultural responses in curated knowledge sources and avoiding unsupported recommendations, particularly for sensitive guidance related to pesticides and fertilizers. |
| PO8  | Individual and Collaborative Team Work | Enables team members to collaborate across modules such as soil-image analysis, crop recommendation, GenAI/RAG, frontend, backend, notifications, testing, and documentation.                                        |
| PO11 | Life-Long Learning                     | Encourages continuous learning in evolving areas such as Generative AI, RAG, computer vision, multimodal AI, and AI applications in agriculture.                                                                     |

**Program Specific Outcomes (PSOs):**

| PSO Number | Name of the PSO Targeted                                                                                                                                                                     | Justification                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                |
|------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| PSO1       | Apply the knowledge of Artificial Intelligence to design, develop, and evaluate computational solutions for complex problems in diverse domains, such as healthcare, finance, and automation | The project applies Artificial Intelligence to address complex agricultural decision-support challenges by integrating soil-image analysis, crop recommendation, personalized crop lifecycle planning, timely notifications, and a Generative AI-powered knowledge assistant. The proposed system involves the design, development, and evaluation of AI-based computational modules intended to provide context-aware agricultural guidance using available soil, crop, weather, and lifecycle information. |
| PSO2       | Demonstrate expertise in using advanced ML tools, techniques, and frameworks to develop innovative solutions for data analysis, pattern recognition, and intelligent decision-making system  | The project utilizes machine learning and deep learning techniques for analyzing soil images and identifying relevant visual patterns, recommendation models for suggesting potentially suitable crops, and Generative AI with Retrieval-Augmented Generation (RAG) for knowledge retrieval and context-aware response generation.                                                                                                                                                                           |

## Course Outcomes (COs):

**CO1** - Apply fundamental and disciplinary concepts and methods in ways appropriate to their principal areas of study.

**CO2** - Demonstrate skill and knowledge of current information and technological tools and techniques specific to the professional field of study.

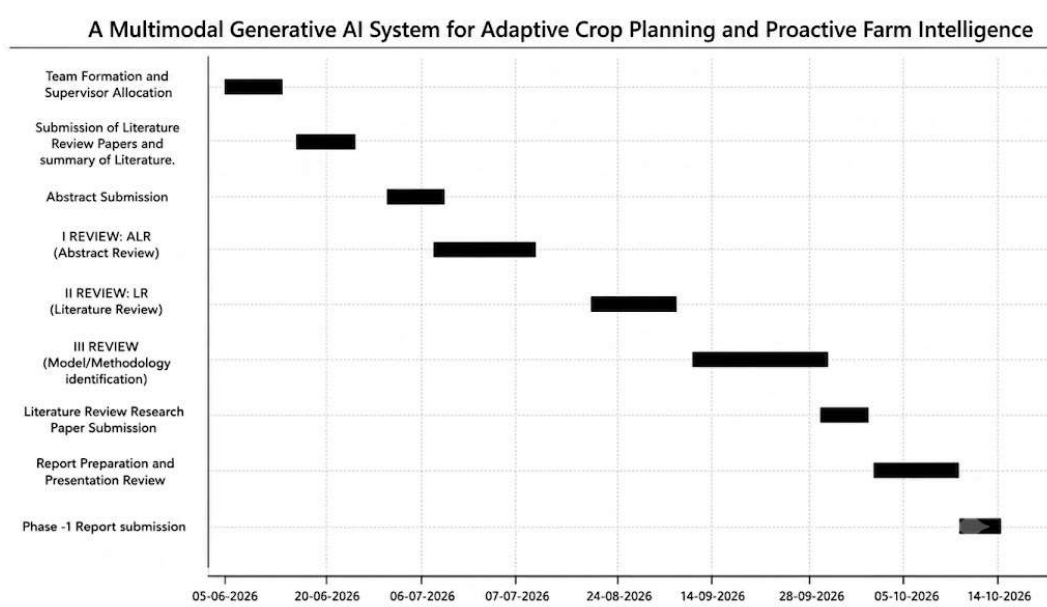
**CO3** - Identify, analyze, and solve problems creatively through sustained critical investigation.

**CO4** - Demonstrate awareness and application of appropriate personal, societal, and professional ethical standards.

**CO5** - Practice the skills, diligence, and commitment to excellence needed to engage in lifelong learning.

**Bloom's Taxonomy Level:** Create

## 11. Timeline – 2026



## 12. References

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Signature of the Supervisor

Signature of Coordinator

Signature of HOD